Rooftop PV Coursework Assessment Rubric

A PV resource and rooftop assessment | 30 marks

Assess the completed practical, independent investigation and combined report. Total: 100 marks. Part A: 30; Part B: 50; Part C: 20. Select forecasting or clustering for Part B.

F Inadequate | D Barely adequate | C Satisfactory | B Good | A Outstanding

Criterion and marks	F	D	C	B	A
City and solar resource 6 marks	City or data are unclear; plots are missing or incorrect.	City and period are identified, but plots contain errors or interpretation is superficial.	Relevant plots are mostly correct; one resource pattern is described.	Correct plots and units support a quantified seasonal or daily comparison.	Well-chosen plots support a precise explanation of resource patterns and their PV implications.
Fixed versus tracking systems 8 marks	POA or energy comparison is absent or fundamentally incorrect.	Comparison is incomplete; units, gain calculation or concepts contain major errors.	Both systems are compared correctly with basic gain calculations and explanation.	POA and energy differences are quantified and explained with solar geometry and operating conditions.	Accurate comparisons across conditions support a well-justified mounting judgment, considering practical constraints.
Rooftop PV potential 10 marks	Roof energy estimates are absent or unsupported.	Some outputs are supplied, but area, units or scaling contain important errors.	Map, regional total, unit-area yield and one building estimate are mostly correct; assumptions are stated.	All required results are correct and interpreted; usable area and footprint overlap are considered.	Transparent calculations and quantified sensitivity support a convincing building recommendation and regional interpretation.
Parameters and shadows 6 marks	Parameter experiment and shadow evidence are missing or incorrect.	A parameter or time is changed with little valid comparison.	A parameter comparison and two shadow times are shown with basic explanations.	Controlled parameter changes are quantified; shadow differences are explained using geometry and solar position.	Clear evidence supports a design judgment; geometric shadow observations and the unshaded energy model are interpreted accurately.

Part A: allocate half of each criterion to calculation and figure quality, and half to interpretation and supported judgment. Apply the descriptors to each half.

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B Independent clustering or forecasting investigation | 50 marks

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Criterion and marks	F	D	C	B	A
Question and data preparation 10 marks	No workable question; data handling invalidates the investigation.	Question is vague; timing, missing data or preprocessing have major errors.	A feasible question and mostly correct features, splits and preprocessing are documented.	A specific question guides appropriate features, chronological evaluation and transparent missing-data handling.	A focused question and justified data choices anticipate sampling, seasonal and information-availability issues.
Initial implementation 10 marks	No usable implementation or results.	Implementation produces partial results with major methodological errors.	Forecasting includes persistence and a lightweight model; or clustering provides groups and representative profiles.	The selected track is implemented correctly with clear settings and appropriate checks.	The implementation is transparent and well justified; outputs demonstrate a sound understanding of the method.
Controlled comparison 10 marks	No meaningful comparison.	Changes are confounded or compared on incompatible data.	A relevant comparison uses mostly consistent data and reports outcomes.	A fair comparison changes a stated factor and uses common periods or samples with suitable measures.	Careful control and quantitative evidence isolate the effect of the change and support a convincing conclusion.
Analysis of results 10 marks	Conclusions are unsupported or contradict results.	Results are described superficially; key measures or examples are missing.	Forecast errors and curves, or group characteristics and quality evidence, receive a basic interpretation.	Numerical evidence and specific difficult or ambiguous cases explain strengths and weaknesses.	Insightful analysis links patterns to PV conditions, examines alternative explanations and bounds the conclusions.
Evidence-led second experiment 10 marks	No second experiment.	A change is made without a clear link to initial evidence.	An observed issue motivates a plausible change; its outcome is reported.	A clear hypothesis follows from the first results and is tested fairly with an explained outcome.	A targeted experiment tests the explanation convincingly and draws a useful conclusion even when performance does not improve.

Use the same criteria for either track. Reward sound reasoning and fair experiments. Model complexity and improved accuracy alone do not earn extra marks.

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C Conclusions and communication | 20 marks

F Inadequate | D Barely adequate | C Satisfactory | B Good | A Outstanding

Criterion and marks	F	D	C	B	A
Integrated conclusions and recommendations 10 marks	No supported conclusion or recommendation.	Conclusions are generic or largely disconnected from the evidence.	Evidence from Parts A and B supports a basic PV-related conclusion; an assumption is identified.	Numerical findings from both parts support a coherent recommendation and explain an important uncertainty.	A persuasive synthesis translates both analyses into a specific, proportionate recommendation and explains what could change it.
Report quality 5 marks	Report is missing or too unclear to assess.	Organization and figure labeling frequently obstruct understanding.	The report is understandable; most figures have labels, units and captions.	A clear question-led structure and readable figures support the reasoning within the page limit.	Concise writing and carefully selected figures make the argument easy to assess; every figure contributes evidence.
Reproducibility and contributions 5 marks	Code or essential files are missing; contributions are unclear.	Substantial manual repair is needed; settings or contributions are poorly documented.	Notebooks and outputs are supplied with basic settings, contribution and AI-use statements.	Notebooks run in order using supplied data; settings, outputs, individual contributions and AI use are clear.	A complete, well-organized submission allows straightforward reproduction and makes individual work and AI assistance transparent.

Marking guidance

Use the descriptors as qualitative anchors and award marks within each criterion maximum. Record a short evidence-based reason for the mark. Letter-to-number conversion and final subject-grade weighting follow the institution's approved policy.

Submission: completed practical and investigation Notebooks with outputs, relevant results, the practical website, and one PDF or PowerPoint report of at most 10 pages/slides including cover, contributions and references. Test labels are supplied for final evaluation after validation-based selection.

Student or group _____ Assessor _____
Part A ____ / 30 Part B ____ / 50 Part C ____ / 20 Total ____ / 100
Feedback _____